Scalable, open-access and multidisciplinary data integration pipeline for climate-sensitive diseases

SUPPLEMENTARY MATERIAL

Abhishek Dasgupta[1,2], Iago Perez-Fernandez[3], Tuyen Huynh[4], Cathal Mills[2,5], Rowan C. Nicholls[1,2], Prathyush Sambaturu[2,6], Marc Choisy[4,7], David Wallom[3], Tung Nguyen Duy[4], Rhys P. D. Inward[2,6], John-Stuart Brittain[1,2], Sarah Sparrow[3], Moritz U.G. Kraemer[2,6]

1. Oxford Research Software Engineering Group, Doctoral Training Centre, University of Oxford, United Kingdom
2. Pandemic Sciences Institute, University of Oxford, United Kingdom
3. Oxford e-Research Centre, Department of Engineering Science, University of Oxford, United Kingdom
4. Oxford University Clinical Research Unit (OUCRU), Ho Chi Minh City, Vietnam
5. Department of Statistics, University of Oxford, United Kingdom
6. Department of Biology, University of Oxford, United Kingdom
7. Nuffield Department of Medicine, University of Oxford, Oxford, United Kingdom

MAIN ARTICLE: https://wellcomeopenresearch.org/articles/10-467

Figure S1. Map of administrative boundaries of the 24 districts (Global Administrative Level 2) of the HCMC province (Global Administrative Level 1), Vietnam, in 2020.

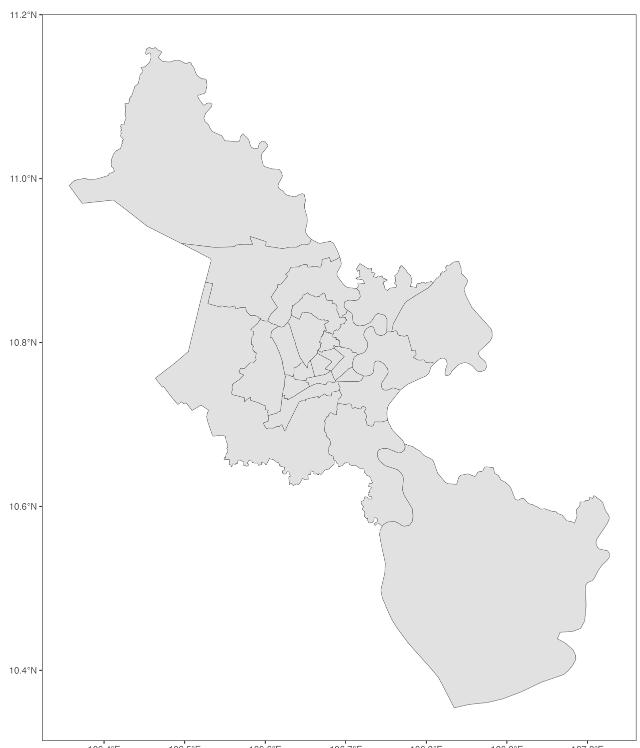


Figure S2. Reported dengue incidence in HCMC, Vietnam, from 2000 to 2022. From 2000 to 2016, only in-patient admissions were recorded, with data collected and validated manually by HCDC. At the start of 2017, all hospitals and clinics in Vietnam began using a new infectious disease incidence reporting portal (*41*) to report case admissions for both in-patient and out-patient. This change resulted in substantial increases in the observed dengue incidence. This rise in reported cases is not necessarily indicative of an unusual outbreak but instead reflects the ability of smaller hospitals and clinics throughout the city to report data to the national system. HCDC collaborates with this system by filtering and downloading incidence reports for patients whose residential addresses are in HCMC. They then manually validate these addresses to confirm that said patients reside in HCMC.

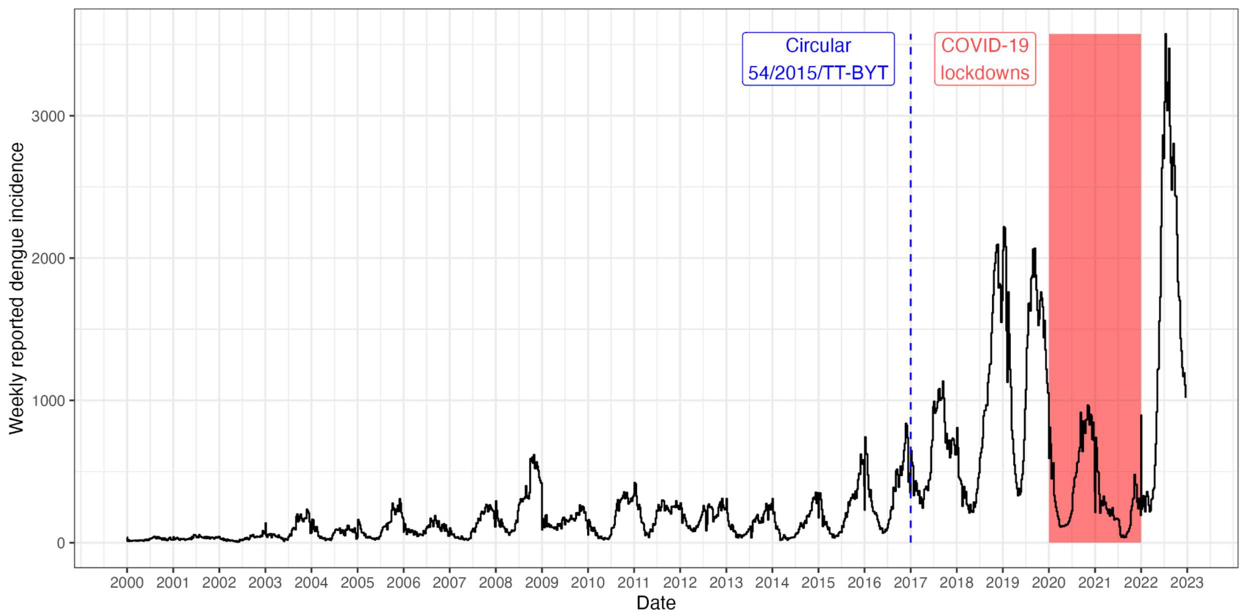
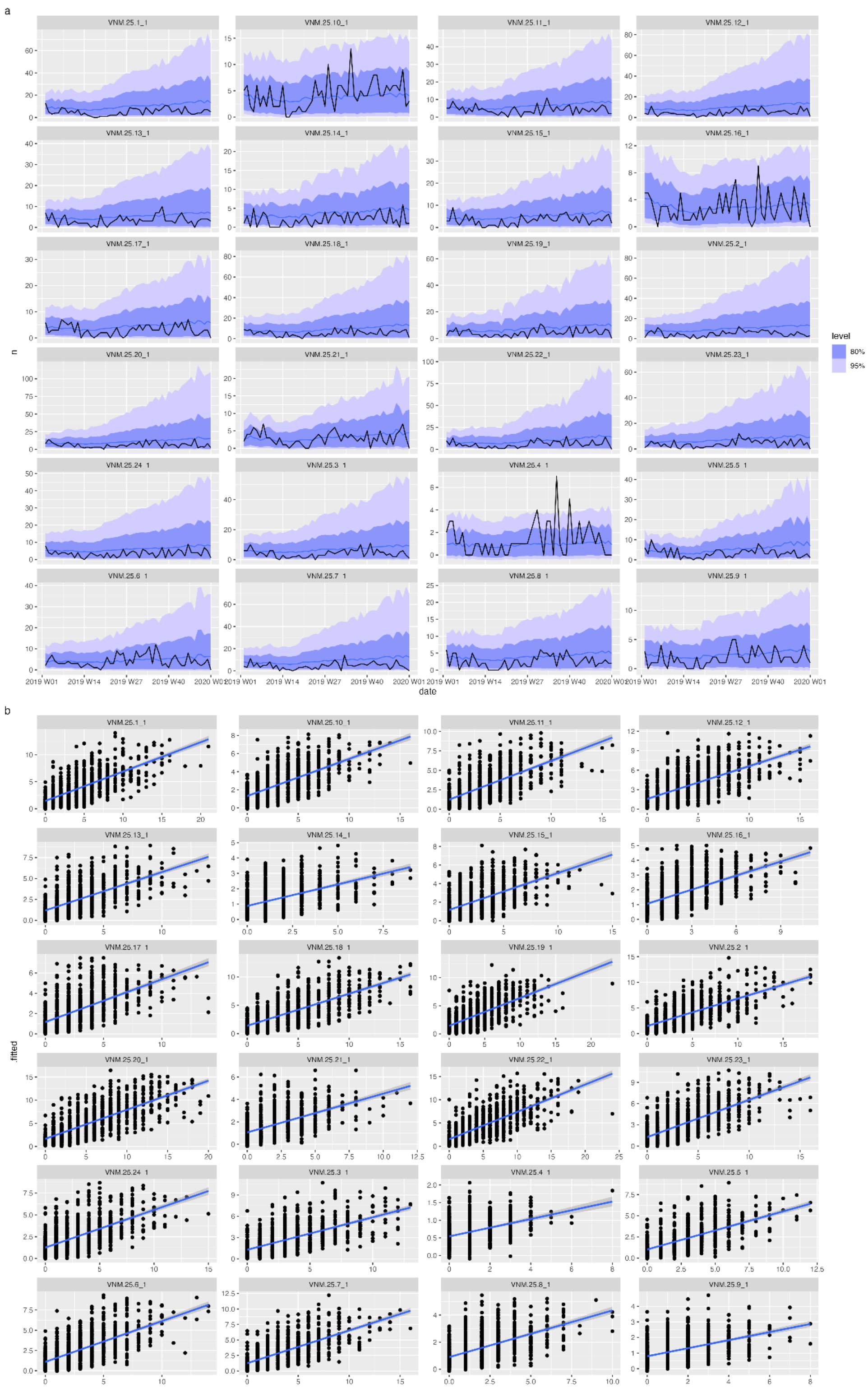


Figure S3. Predicted dengue incidence from a SARIMAX model trained using synthetic case incidence data, and ERA5 weather data zonally aggregated from 2001-2019, together with 2-week ECMWF forecast from 11 July 2025. Figure S3(a) shows the forecast timeseries and Figure S3(b) shows the R^2 values for administrative level 2 regions in Ho Chi Minh City.



Supplementary tables

Table S1. For each dataset and variable methods to resample them to a common resolution and aggregation matrices relevant for dengue modelling. Derived variables are calculated after resampling, but before spatial aggregation. All datasets are licensed under CC-BY-4.0 (Creative Commons Attribution 4.0) unless otherwise stated.

<table>
<tr><th>Dataset Source</th><th>Variable</th><th>Resampling method (to ~1km at equator, 30-arc seconds)</th><th>Spatial aggregation</th><th>Temporal aggregation</th><th>Source</th></tr>
<tr><td colspan="6">Meteorological</td></tr>
<tr><td rowspan="5">ERA5 atmospheric reanalysis (era5)</td><td>Mean/max/min 2m temperature (Kelvin)

t2m, mn2t24, mx2t24</td><td>Bilinear interpolation (24)</td><td>Population and area weighted mean of daily mean, mean of daily max, mean of daily min</td><td>Weekly mean</td><td rowspan="2">(42)</td></tr>
<tr><td>Total daily precipitation (m)
tp</td><td>Distance-weighted average remapping (25)</td><td>Population and area weighted sum</td><td>Weekly sum</td></tr>
<tr><td>Mean/max/min relative humidity (%)

r, mxr24, mnr24</td><td rowspan="2">Bilinear interpolation (24)</td><td>Population and area weighted mean of daily mean, mean of daily max, mean of daily min</td><td>Weekly mean</td><td rowspan="2">Computed using 2m temperature, 2 metre dew point temperature and surface pressure from (43)</td></tr>
<tr><td>Mean/max/min specific humidity (g/kg)

q, mxq24, mnq24</td><td>Population and area weighted mean of daily mean, mean of daily max, mean of daily min</td><td>Weekly mean</td></tr>
<tr><td>Hydrological</td><td>Distance-weig</td><td>Population</td><td>Weekly sum</td><td>Computed using the</td></tr>
</table>

	balance (m) hb	hted average remapping (25)	and area weighted sum		sum of daily Evaporation plus total precipitation from (*43*)
	Standardised Precipitation Index (units of standard precipitation) measured in time window (rolling average) spi		Population and area weighted sum	Weekly sum	Computed using daily total precipitation from (*43*) and following (*44*)
	Standardised Precipitation-Evaporation Index (units of standard precipitation) spei		Population and area weighted sum	Weekly sum	Computed using daily total precipitation and potential evaporation from (*43*) and following (*45*, *46*)
	Corrected total daily precipitation (m) tp_bc		Population and area weighted sum	Weekly sum	Same as *Total daily precipitation*, but after applying a bias-correction technique (See Methods section)
	Standardised Precipitation Index using corrected total daily precipitation (units of standard precipitation) spi_bc		Population and area weighted sum	Weekly sum	Same as *Standardised Precipitation Index*
	Standardised Precipitation-Evaporation Index using		Population and area weighted sum	Weekly sum	Same as *Standardised Precipitation-Evapo transpiration Index*

	corrected total daily precipitation (units of standard precipitation) spei_bc				
	Hydrological balance using corrected total daily precipitation (m) hb_bc		Population and area weighted sum	Weekly sum	Same as *Daily hydrological balance* but using *corrected total daily precipitation*
ECMWF extended weather forecast data 2 weeks ahead (*47*)	Same variables as ERA5, except hydrological balance.	See above for details for each variable	See above for details for each variable	See above for details for each variable	
Socio-demographic					
WorldPop 2000-2030 2021-2030 are predicted estimates	Population counts pop	NA (already at desired target resolution)	Area weighted sum	Yearly	2000-2020: (*48*) 2021-2030: (*49*)
Meta Creative Commons Attribution-Non Commercial 4.0 International (CC BY-NC 4.0)	Relative wealth index rwi	Available at 2.4km resolution	Population weighted mean	Static (last uploaded April 8, 2021)	(*50*)

Table S2. Formulae for spatial aggregation used by *exactextract (51)*. Here x_i is the weather raster at pixel *i* resampled to be of the same resolution as the population raster, w_i is the population raster, and c_i is the coverage weight, which is the fraction of the polygon covered by both weather and population rasters multiplied by the spherical area approximation of pixel *i* in km^2.

Aggregation type	Formula
weighted_mean	$\frac{\Sigma x_i c_i w_i}{\Sigma c_i w_i}$
area_weighted_sum	$\frac{\Sigma x_i c_i w_i}{\Sigma c_i}$
weighted_variance	$\frac{\Sigma c_i w_i (x_i - \bar{x})}{\Sigma c_i w_i}$
mean	$\frac{\Sigma x_i c_i}{\Sigma c_i}$
sum	$\Sigma x_i c_i$
count	Σc_i
weighted_sum	$\Sigma x_i c_i w_i$

Table S3. Automated validation checks by variable. We conduct automated range checks for a subset of variables in addition to ensuring that no variable has an NA value.

Metric	**Range**
era5.2m_temperature	175 — 325 K (-98.15 — 61.85 ºC) These are slightly wider than the actual lowest and highest air temperatures recorded
era5.total_precipitation	0 — (no upper bound defined)
era5.wind_speed	0 — 110 m/s
era5.relative_humidity	0 — 100 (%)
era5.specific_humidity	0 — (no upper bound defined)